SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS **COURSE CODE: MLA01**

COURSE OUTCOME COVERED

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems.* (BTL 3)

Assessment Tool Weightage: 55 %

Total Marks: 50 **Time: 60 Mins**

No. of Questions: 1

Annexure A

Agent Design Assessment

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Question	Problem Understanding & AI Concept Identification	Dialogflow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

Q. No.	Criteria to be evaluated	BL	Marks
1	**Problem Analysis and Agent Design:** Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10
2	**Dialogflow Agent Development:** Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10
3	**Problem Formulation and Decision Flow:** Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	**Search Strategy Application:** Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10
5	**Testing and Evaluation:** Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

MLA0103-Artificial Intelligence with Expert Systems

Assesement Tool

Name: Tejaswini Reg no: 192525171

AI-Based Student Support Assistant using Google Dialogflow ES

Table of Contents:

Problem Statement

Universities receive a large number of repetitive queries from students regarding course registration, class timetables, examination schedules, attendance details, library timings, hostel facilities, and other campus services. Students often have to contact faculty members or administrative staff to obtain this information, which can result in delays, increased workload for staff, and inconvenience for students. To address this problem, an **AI-based Student Support Assistant** is developed using **Google Dialogflow ES**. The chatbot provides instant, accurate, and automated responses to common student queries at any time, thereby improving communication and reducing administrative effort.

Agent Name: Student Assistant

Type of Agent: Conversational AI chatbox

Google Dialogflow:

Google Dialogflow ES uses Natural Language Understanding (NLU) powered by Google's machine learning models. It identifies user intent and extracts entities from natural language queries to generate appropriate responses.

Environment:

The chatbot operates in an online environment using Google Dialogflow ES. It can later be integrated into university websites, mobile applications, or messaging platforms.

Inputs

The chatbot accepts text-based queries such as:

- How do I register for courses?
- Show my timetable.
- When are the semester exams?
- What is my attendance percentage?
- What are the library timings?

Outputs

The chatbot provides appropriate text responses such as:

- Course registration procedure
- Timetable information
- Examination schedule
- Attendance details
- Campus service information

Sensors

The chatbot receives user input through:

- Text entered by the student in the chat interface

Objectives:

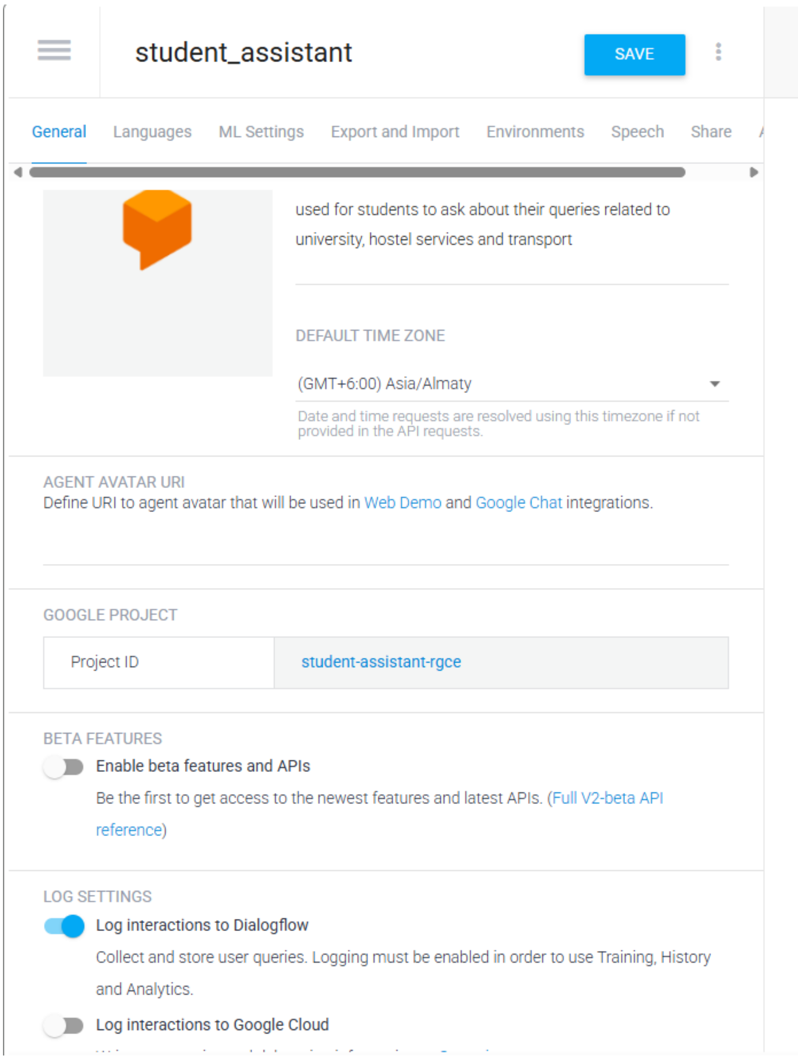
The main objectives of the Student Support Assistant are:

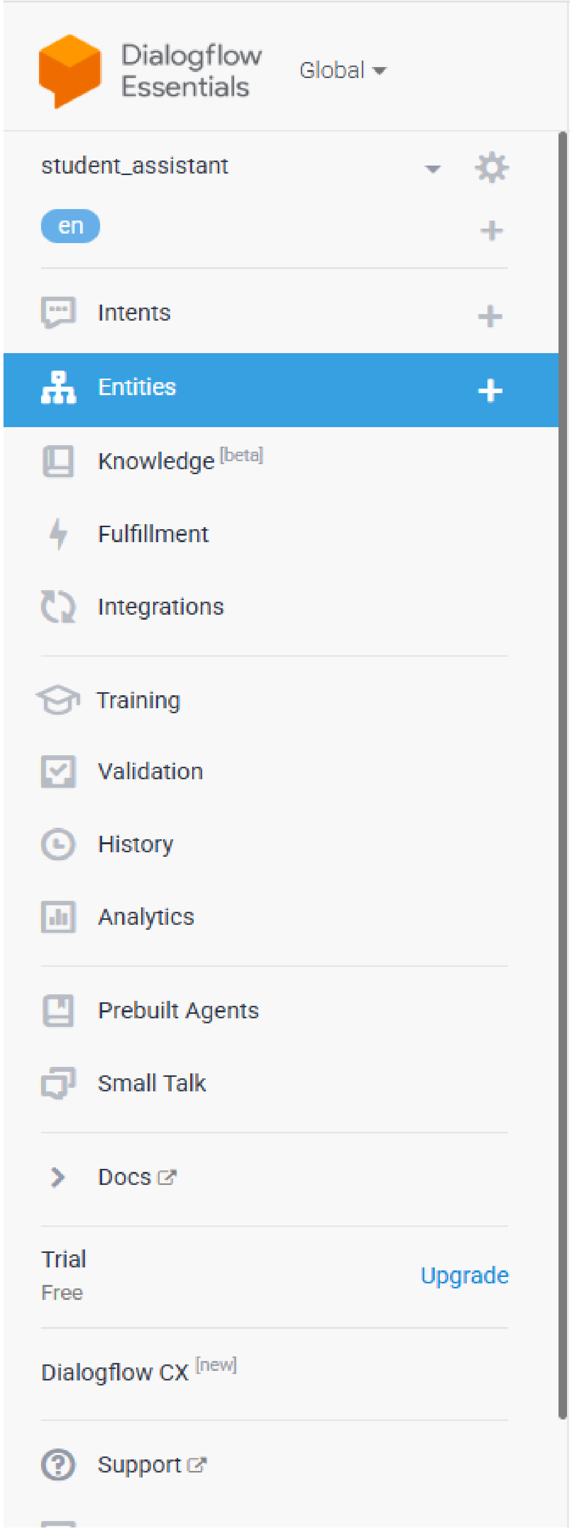
- To provide instant responses to student queries.
- To automate frequently asked questions related to academic services.
- To reduce the workload of university administrative staff.
- To improve the availability of student support services.
- To provide accurate and consistent information 24×7.

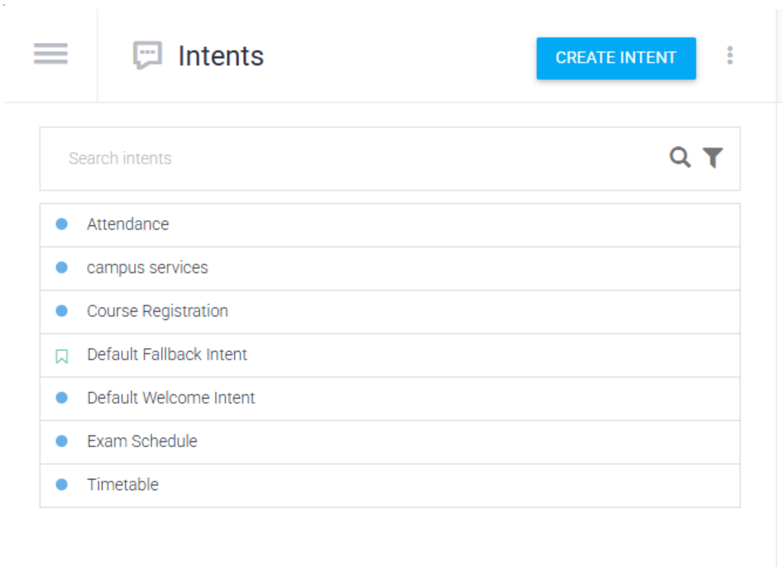
Knowledge Base

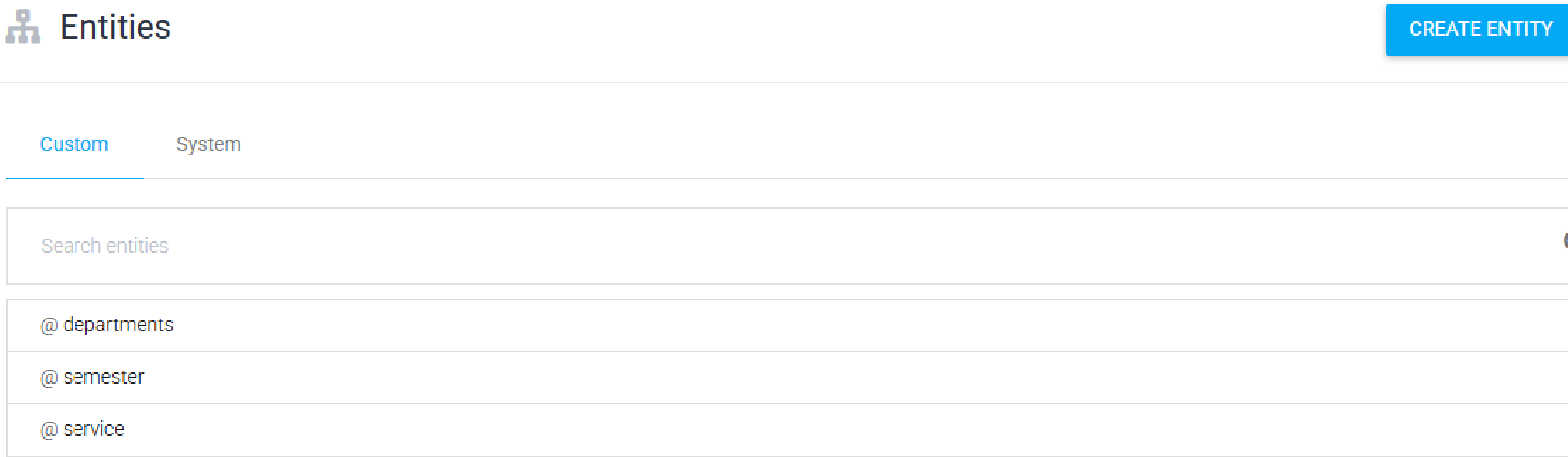
The chatbot stores information related to:

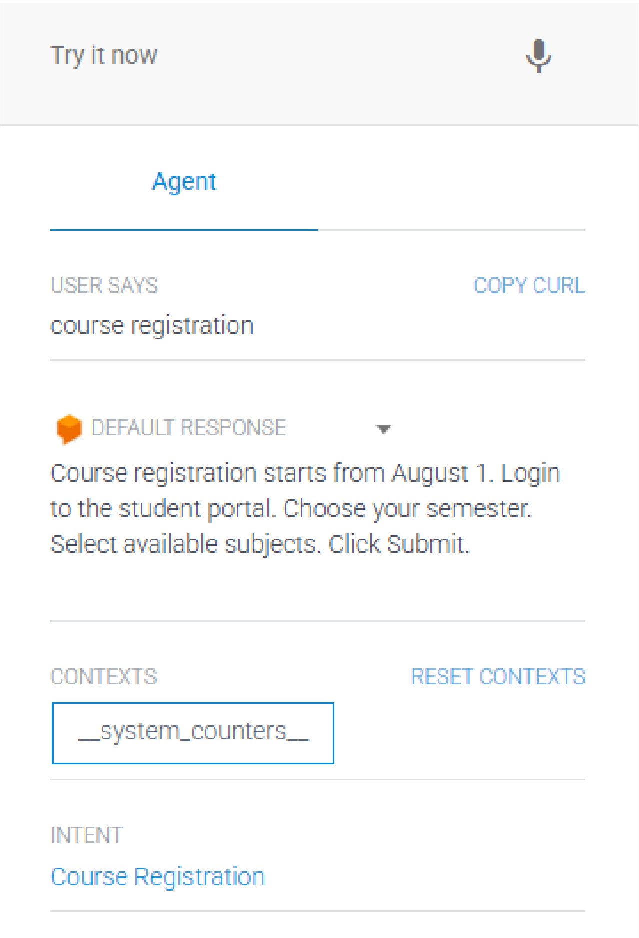
- Course Registration
- Timetable
- Examination Schedule
- Attendance
- Library Services
- Hostel Information
- Medical Centre
- Transport Services

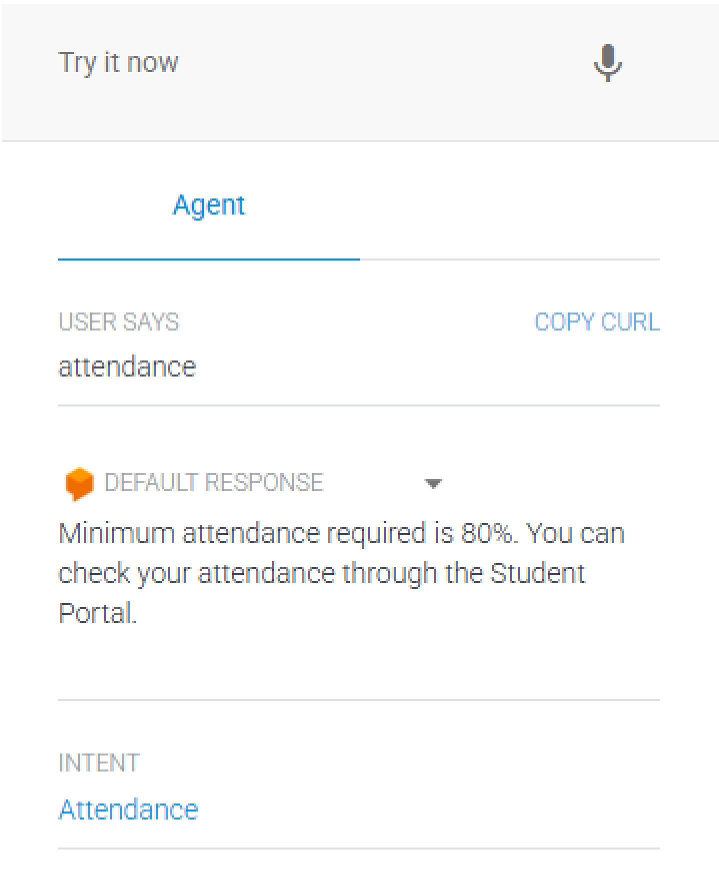

student_assistant
SAVE
General
Languages
ML Settings
Export and Import
Environments
Speech
Share
used for students to ask about their queries related to university, hostel services and transport
DEFAULT TIME ZONE
(GMT+6:00) Asia/Almaty
Date and time requests are resolved using this timezone if not provided in the API requests.
AGENT AVATAR URI
Define URI to agent avatar that will be used in Web Demo and Google Chat integrations.
GOOGLE PROJECT
Project ID
student-assistant-rgce
BETA FEATURES
Enable beta features and APIs
Be the first to get access to the newest features and latest APIs. (Full V2-beta API reference)
LOG SETTINGS
Log interactions to Dialogflow
Collect and store user queries. Logging must be enabled in order to use Training, History and Analytics.
Log interactions to Google Cloud

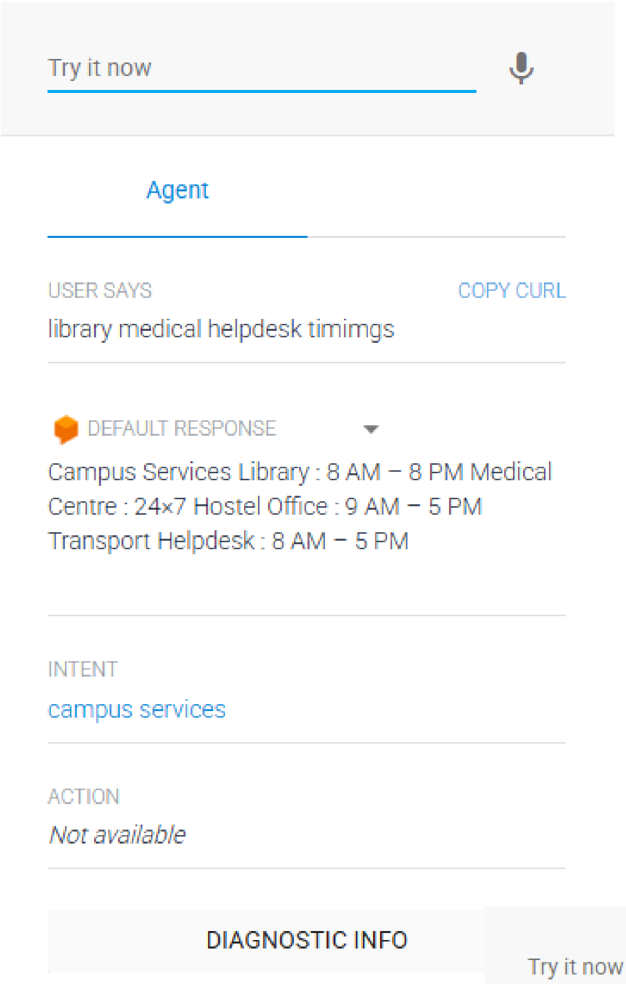

Dialogflow
Essentials
Global
student_assistant
en
Intents
Entities
Knowledge [beta]
Fulfillment
Integrations
Training
Validation
History
Analytics
Prebuilt Agents
Small Talk
Docs
Trial
Free
Upgrade
Dialogflow CX [new]
Support

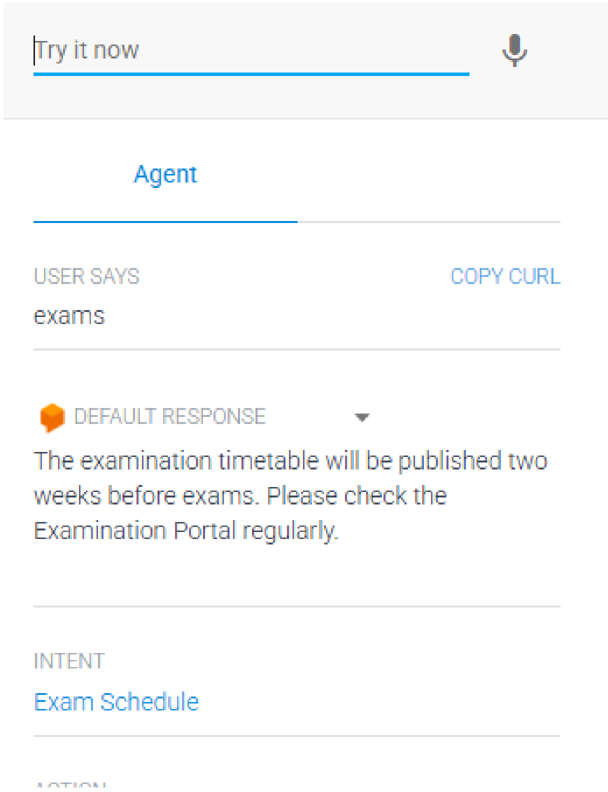

Intents
CREATE INTENT
Search intents
Attendance
campus services
Course Registration
Default Fallback Intent
Default Welcome Intent
Exam Schedule
Timetable

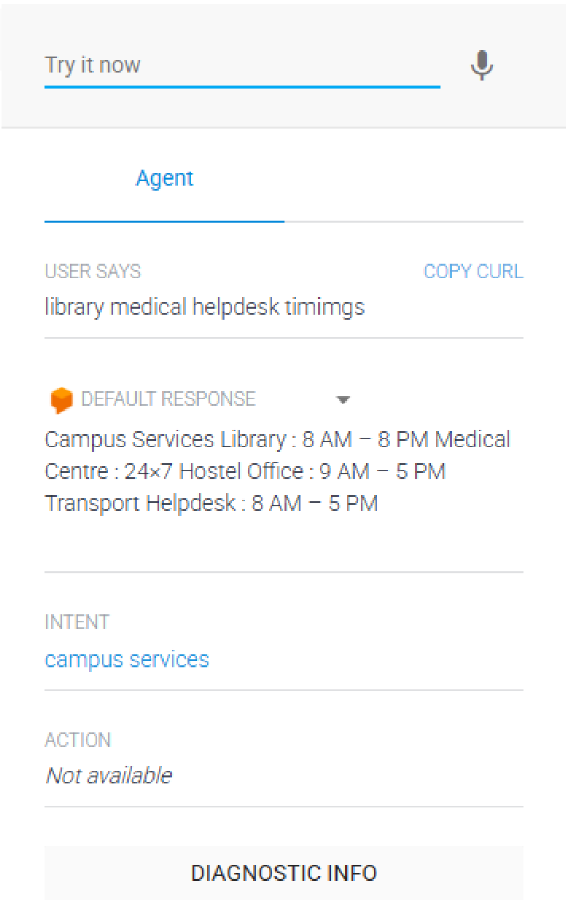

Entities
CREATE ENTITY
Custom
System
Search entities
@ departments
@ semester
@ service

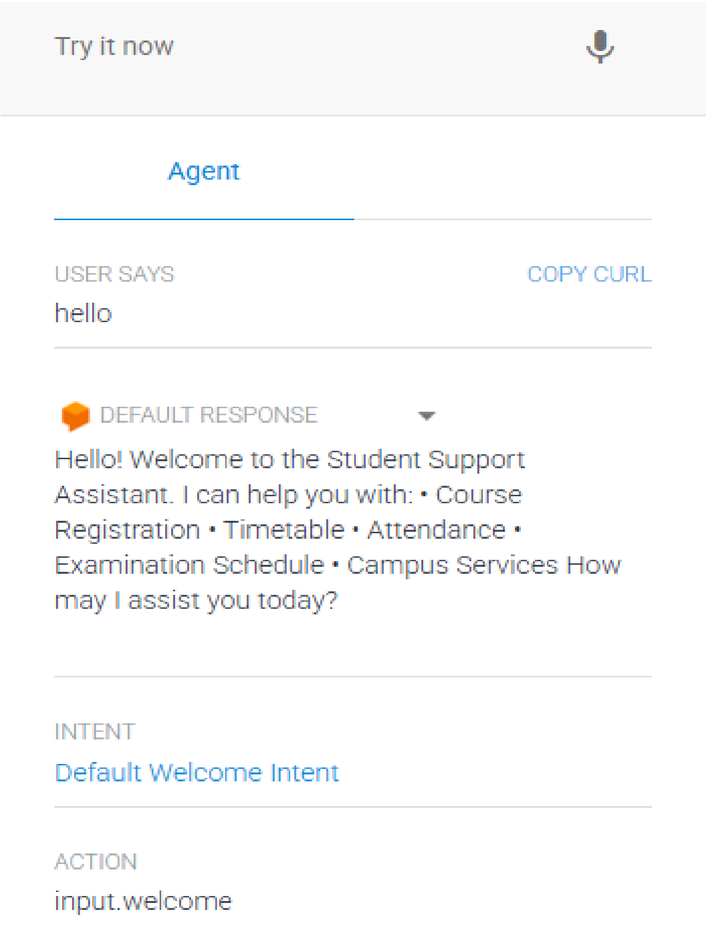

Try it now
Agent
USER SAYS
COPY CURL
course registration
DEFAULT RESPONSE
Course registration starts from August 1. Login to the student portal. Choose your semester. Select available subjects. Click Submit.
CONTEXTS
RESET CONTEXTS
__system_counters__
INTENT
Course Registration

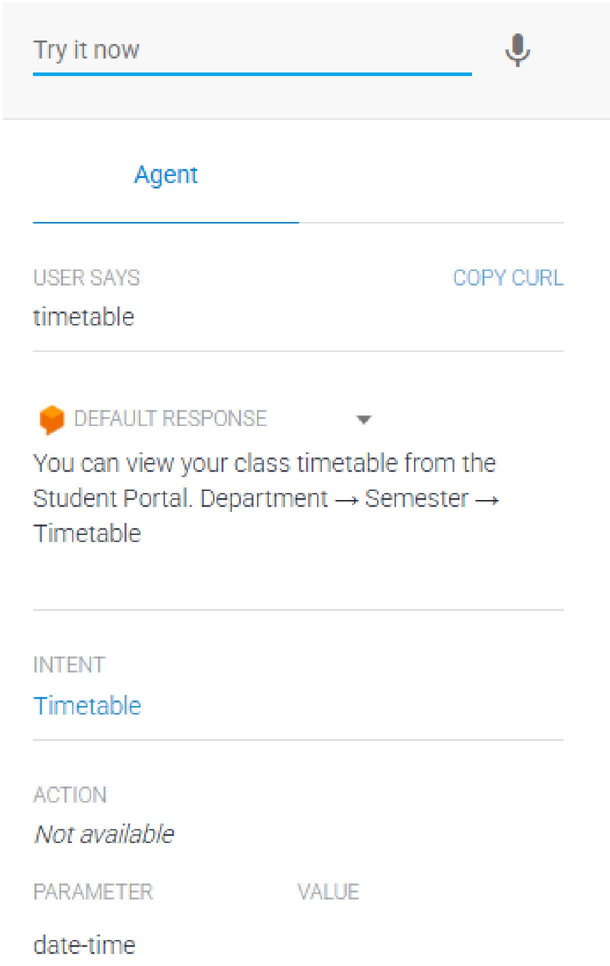

Try it now
Agent
USER SAYS
COPY CURL
attendance
DEFAULT RESPONSE
Minimum attendance required is 80%. You can check your attendance through the Student Portal.
INTENT
Attendance

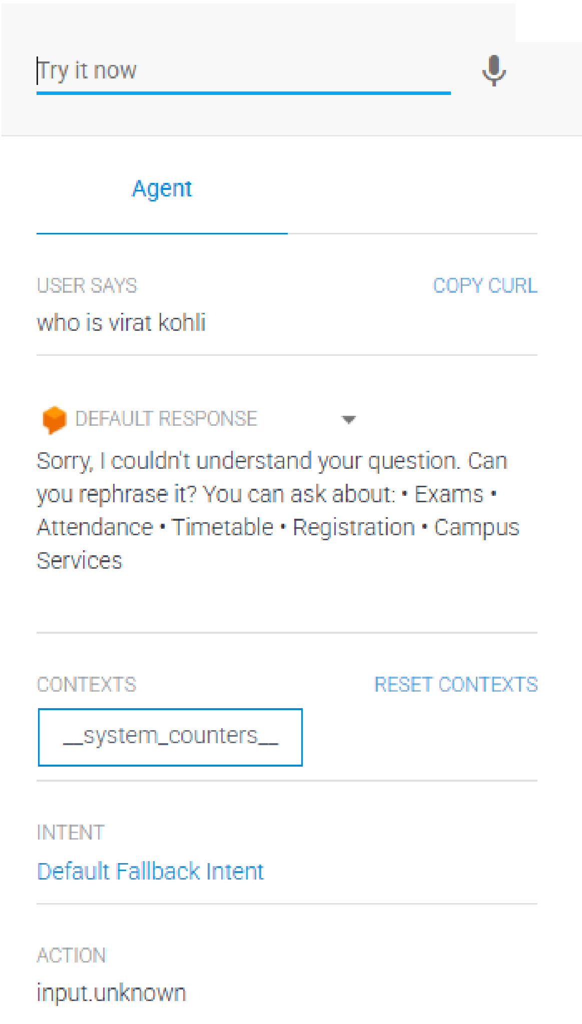

Try it now
Agent
USER SAYS
COPY CURL
library medical helpdesk timimgs
DEFAULT RESPONSE
Campus Services Library : 8 AM – 8 PM Medical Centre : 24×7 Hostel Office : 9 AM – 5 PM Transport Helpdesk : 8 AM – 5 PM
INTENT
campus services
ACTION
Not available
DIAGNOSTIC INFO

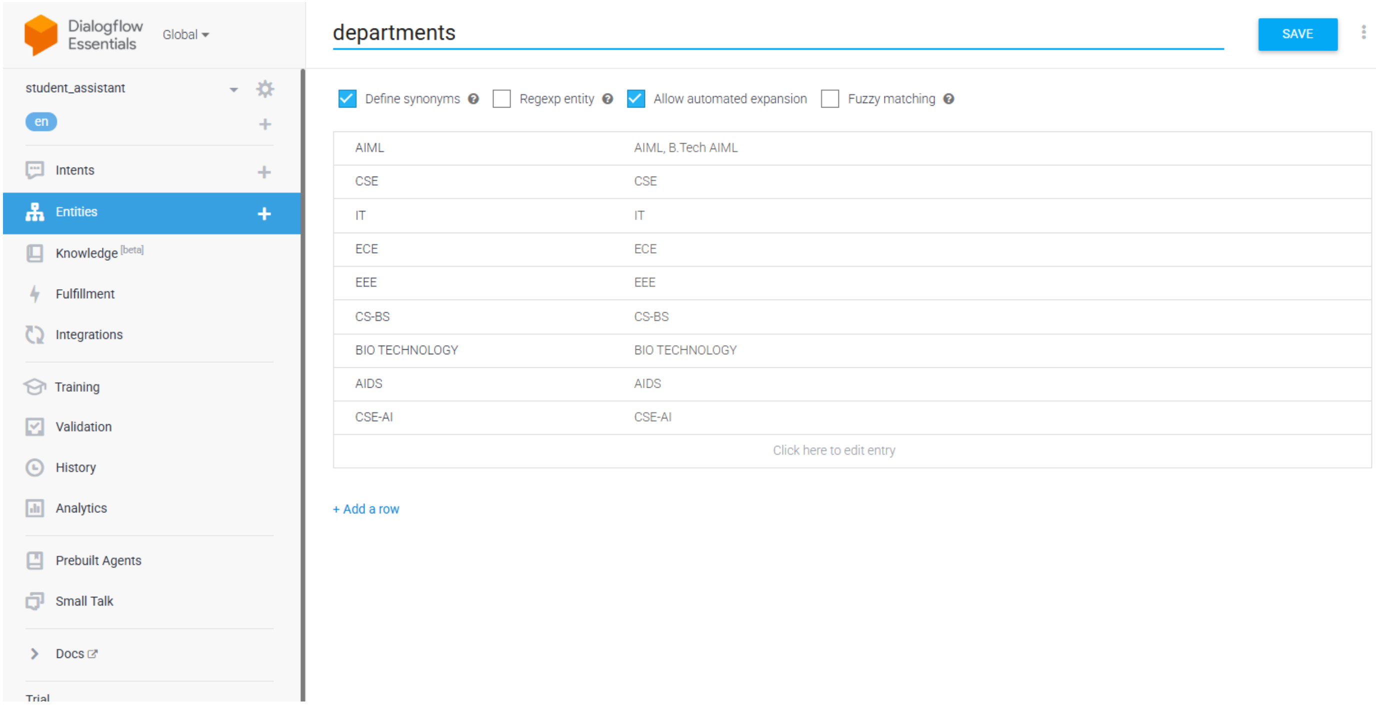

Try it now
Agent
USER SAYS
COPY CURL
exams
DEFAULT RESPONSE
The examination timetable will be published two weeks before exams. Please check the Examination Portal regularly.
INTENT
Exam Schedule


Try it now
Agent
USER SAYS
COPY CURL
library medical helpdesk timimgs
DEFAULT RESPONSE
Campus Services Library : 8 AM – 8 PM Medical Centre : 24×7 Hostel Office : 9 AM – 5 PM Transport Helpdesk : 8 AM – 5 PM
INTENT
campus services
ACTION
Not available
DIAGNOSTIC INFO

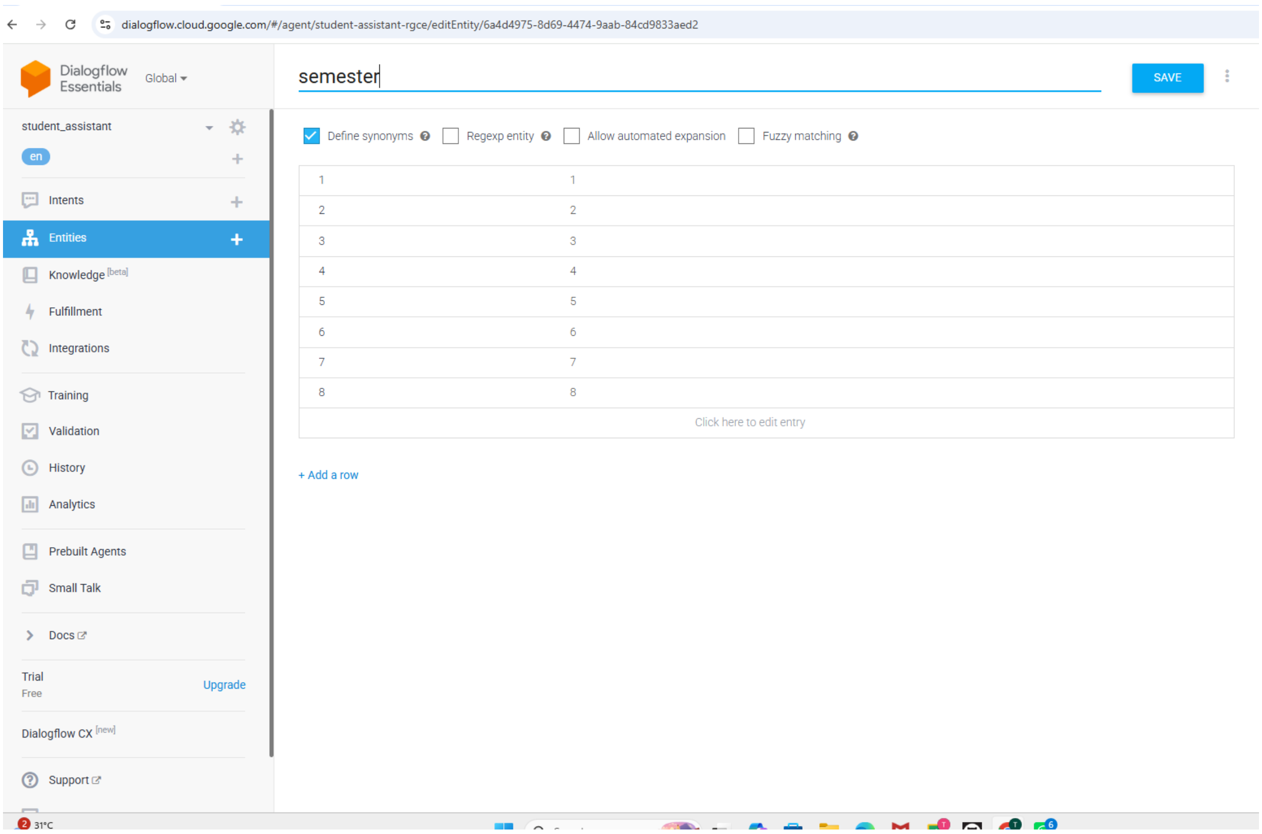

Try it now
Agent
USER SAYS
COPY CURL
hello
DEFAULT RESPONSE
Hello! Welcome to the Student Support Assistant. I can help you with: • Course Registration • Timetable • Attendance • Examination Schedule • Campus Services How may I assist you today?
INTENT
Default Welcome Intent
ACTION
input.welcome


Try it now
Agent
USER SAYS
COPY CURL
timetable
DEFAULT RESPONSE
You can view your class timetable from the Student Portal. Department → Semester → Timetable
INTENT
Timetable
ACTION
Not available
PARAMETER
VALUE
date-time


Try it now
Agent
USER SAYS
COPY CURL
who is virat kohli
DEFAULT RESPONSE
Sorry, I couldn't understand your question. Can you rephrase it? You can ask about: • Exams • Attendance • Timetable • Registration • Campus Services
CONTEXTS
RESET CONTEXTS
__system_counters__
INTENT
Default Fallback Intent
ACTION
input.unknown



Dialogflow Essentials
Global
student_assistant
en
Intents
Entities
Knowledge [beta]
Fulfillment
Integrations
Training
Validation
History
Analytics
Prebuilt Agents
Small Talk
Docs
departments
SAVE
Define synonyms
Regexp entity
Allow automated expansion
Fuzzy matching
AIML
AIML, B.Tech AIML
CSE
CSE
IT
IT
ECE
ECE
EEE
EEE
CS-BS
CS-BS
BIO TECHNOLOGY
BIO TECHNOLOGY
AIDS
AIDS
CSE-AI
CSE-AI
Click here to edit entry
+ Add a row



dialogflow.cloud.google.com/#/agent/student-assistant-rgce/editEntity/6a4d4975-8d69-4474-9aab-84cd9833aed2
Dialogflow Essentials
Global
student_assistant
en
Intents
Entities
Knowledge [beta]
Fulfillment
Integrations
Training
Validation
History
Analytics
Prebuilt Agents
Small Talk
Docs
Trial
Free
Upgrade
Dialogflow CX [new]
Support
semester
SAVE
Define synonyms
Regexp entity
Allow automated expansion
Fuzzy matching
1
1
2
2
3
3
4
4
5
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6
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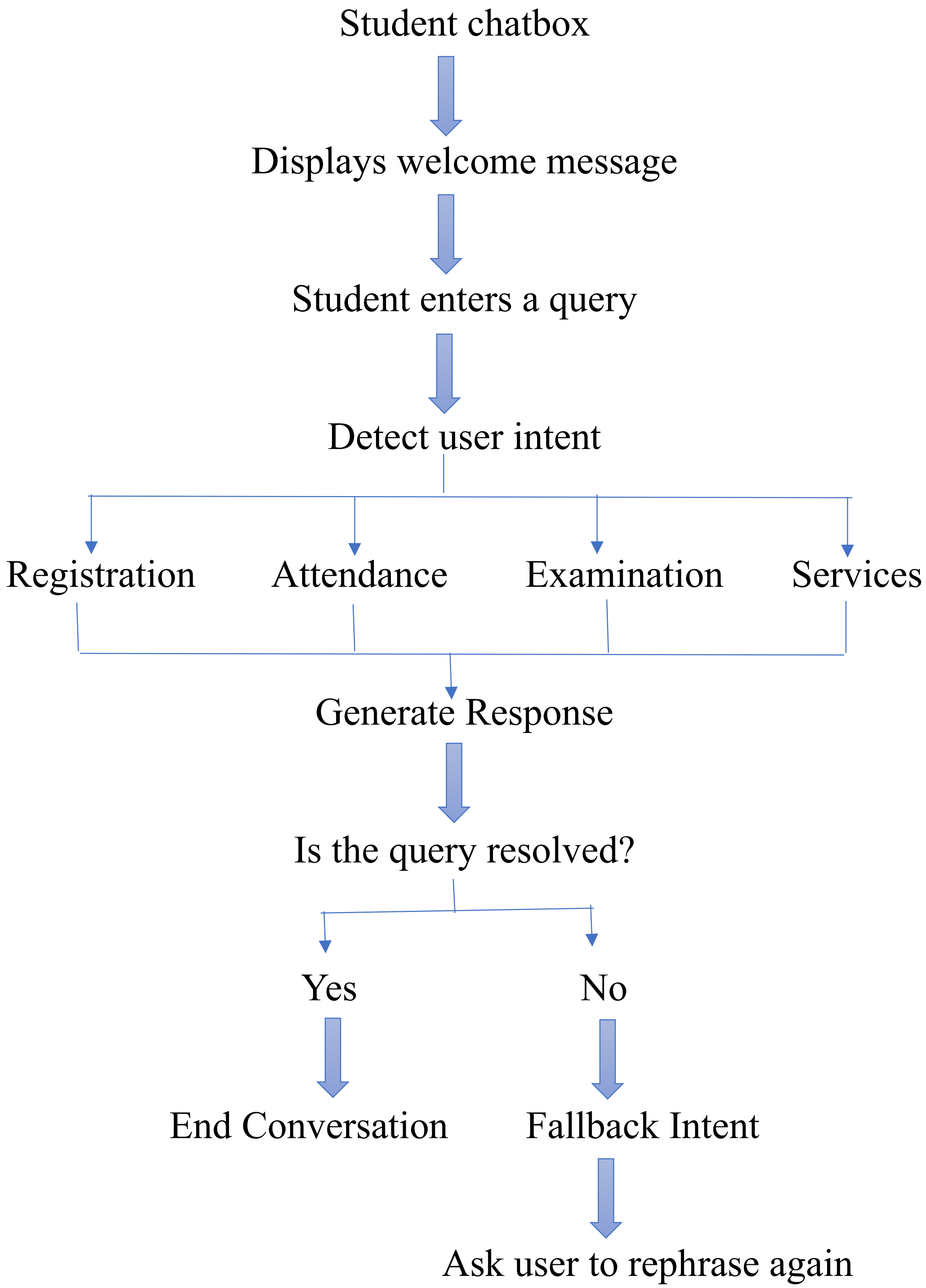
7
7
8
8
Click here to edit entry
+ Add a row
31°C

Conversation Flow Diagram:



Selected Search Strategy

Heuristic-Based Search:

For this project, **Heuristic-Based Search** is selected because Dialogflow ES uses **Natural Language Understanding (NLU)** and **Machine Learning** to compare the user's query with the available training phrases and identify the most relevant intent.

Instead of checking every intent one by one, the chatbot estimates the best match using confidence scores.

Why Heuristic Search?

Heuristic search is suitable for conversational AI because it:

- Quickly identifies the best matching intent.
- Reduces response time.
- Improves chatbot accuracy.
- Efficiently handles large numbers of intents.
- Provides a better user experience.

Since Dialogflow ES uses Google's machine learning models, heuristic search is naturally integrated into the intent-matching process.

Advantages of the Developed Chatbot

- Provides 24×7 student support.
- Reduces administrative workload.
- Delivers quick and accurate responses.
- Improves communication between students and the university.
- Enhances the overall student experience.

Conclusion:

The **AI-Based Student Support Assistant** was successfully tested using multiple student queries. The chatbot accurately recognized user intents, generated relevant responses, and handled unknown queries using the Default Fallback Intent. The evaluation demonstrated high intent recognition accuracy, fast response time, and reliable performance. With future enhancements such as database integration and multilingual support, the chatbot can become a comprehensive digital assistant for university students.

Evaluation Rubrics

Criteria	Weightage (%)	Excellent (Full Marks)	Good	Average	Poor
Problem Understanding & AI Concept Identification	20%	10 – Clearly analyzes the student support problem, defines AI application, agent objective, users, environment, inputs, outputs, and performance measures with proper justification.	7.5 – Identifies most components of the agent design with minor omissions.	5 – Provides basic problem analysis with limited explanation of agent components.	0–2.5 – Incomplete understanding of problem and agent design.
Dialogflow Agent Design & Implementation	30%	10 – Successfully creates a functional Dialogflow agent with appropriate intents, entities, training phrases, responses, contexts, and explains interaction flow with screenshots.	7.5 – Develops major Dialogflow components with minor configuration errors.	5 – Creates basic intents and responses with limited entity/context utilization.	0–2.5 – Incomplete or incorrect Dialogflow implementation.
Problem Formulation & Conversation Flow Design	20%	10 – Clearly represents chatbot interaction using initial state, goal state, actions, state transitions, goal conditions, and a well-designed conversation flow diagram.	7.5 – Provides a structured flow with minor missing details.	5 – Provides basic flow representation with limited explanation.	0–2.5 – Poor or missing problem formulation and decision flow.
Application of AI Techniques & Reasoning	20%	10 – Correctly analyzes and applies suitable search strategies (BFS/heuristic search) for intent selection or dialogue navigation with strong justification.	7.5 – Applies appropriate search strategy with reasonable explanation.	5 – Provides basic understanding of search strategy application.	0–2.5 – Unable to relate search concepts with conversational agent.
Testing, Evaluation & Documentation	10%	10 – Performs comprehensive testing using multiple user queries, evaluates accuracy, response quality, unknown query handling, and suggests meaningful improvements.	7.5 – Performs adequate testing and provides evaluation with minor gaps.	5 – Conducts limited testing with basic observations.	0–2.5 – Insufficient testing and evaluation.